

Log-Based Recovery & Checkpointing DBMS

> **Definition:** Log-Based Recovery maintains a chronological record of all database modifications on non-volatile storage (the **Log**) to restore database Atomicity and Durability following system crashes.

1. Detailed Technical Explanation

1. Write-Ahead Logging (WAL) Protocol

Before any data modification is written to disk, the corresponding log record **MUST** be flushed to non-volatile log storage.

Log Record Formats:

2. Recovery Techniques

1. Deferred Update Technique (NO-UNDO / REDO Algorithm)

- Database disk writes are postponed until the transaction reaches its `COMMIT` point.

2. Immediate Update Technique (UNDO / REDO Algorithm)

- Database disk writes can occur while the transaction is still active.

3. Checkpointing

Checkpointing reduces recovery overhead by periodically flushing all dirty buffer blocks and active transaction logs to disk.

4. Core Concepts & Memory Keywords

- **WAL Protocol:** Write log to disk BEFORE writing database blocks.

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5. Must-Write Points for Exams

- WAL guarantees Durability by ensuring committed transactions can be reconstructed via REDO log records.

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6. Quick Recall Flow

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